Thermal control system

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1 Introduction

The role of the thermal control system is to keep the temperature of the spacecraft components within required temperature limits for given orbits, power demand, operations, and other considerations. The thermal control system should also reduce temperature gradients across the spacecraft and some components like lenses. Two temperature limits are typically defined when considering temperature control of thermal systems: an *operational limit* and a *survival limit*. An effective thermal control system will keep the component temperatures within their operational limits. A component should not lose operability within the survival limit even at extreme temperatures. The survival limit typically $\pm 10°C$ is wider than the operational limit. The following sections describe the workflow required to design a thermal control system. It discusses the design process and hardware systems that are currently available to CubeSat developers. An application of the process to design a CubeSat thermal control system is also demonstrated.

A spacecraft's heat rejection capability is directly related to its size due to the fact that heat is rejected into space only by radiation. Powerful spacecraft should have large radiators to have a significant cooling capability. The same dependency of cooling capability upon unit dimensions is demonstrated by a computer workstation. Computer workstation enclosure size is determined mostly by fan diameter, which is required to remove generated heat. The similarity of two different examples from different industries confirms the fact that cooling capabilities play a significant role in design (Fig. 1).

As we see later, in Section 5, radiator sizing should be included into the spacecraft design process. If radiator size does not match the required waste energy rejection, the spacecraft temperature will increase (or decrease if a radiator is too big) and can exceed an upper limit (or a lower limit if a radiator is too big) of survivable level. This condition could lead to mission failure.

Thermal control technology for CubeSats is primarily based on passive methods, that is, surface finishes, radiators, and, sometimes, heat pipes. The active methods like liquid pumps are not typically used in CubeSats at this moment.

It is not widely recognized that thermal management for CubeSats is more complicated than for large spacecraft like GEO communication satellites. As an example, Table 1 compares a thermal situation in two satellites, GEO communication and 3U CubeSat under following assumptions:

- Only two sides (North and South) are used for heat rejection.
- Power generations are 5000 W for GEO and 50 W 3U CubeSat correspondingly.
- Electronics efficiency is 60% on both satellites, that is, 40% of generated energy is converted into waste heat.

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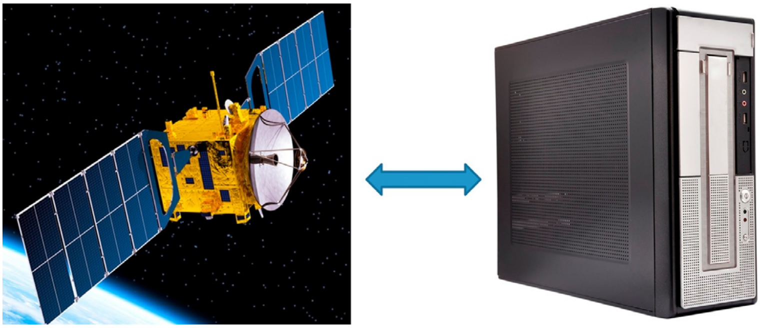

Fig. 1 Satellite versus computer—similar problem and solutions [1]. Reproduced with permission from iStock.

Table 1 Comparison of GEO com and 3U CubeSat satellites.

	GEO com	3U cube sat	Ratio GEO/cube
Satellite dimension [m]	$3.0 \times 3.0 \times 6.0$	$0.1 \times 0.1 \times 0.3$	
Satellite volume [m^3]	54.0	0.003	18,000
Radiator surface [m^2][a]	42.88	0.06	715
Power [W]	5000	50	100
Power density [W/m^3]	92.6	16,666.7	0.6%
Efficiency	60%	60%	
Waste heat	2000	20	100
Rejected heat flux [W/m^2]	46.6	333.3	0.14

[a] Only two largest surfaces are used as radiators.

The data presented in Table 1 clearly shows that thermal management in CubeSat is much more complicated than for large satellites where power density is much lower and the required flux for waste heat rejection is less. This means that thermal control is less complicated in large satellites than for CubeSat due to the large volume and surfaces of GEO satellites available to radiate waste heat.

2 Workflow to design thermal control system for SmallSat spacecraft/CubeSats

Historically, designs of CubeSat thermal systems either never performed methodically or performed in a sporadic manner due to the incorrect opinion that a thermal control system is not important. Descriptions of often limited approaches to CubeSat thermal management systems and related hardware can be found in the existing literature [2].

Previously, CubeSats had been built mostly by universities with very low power levels, around 1–5 W. As we will see in the following sections, the current demand for CubeSats with 50 W or more changes completely the thermal requirements and, therefore, the approach to the spacecraft design. An effective thermal control system must be included in the design of the CubeSat to prevent failure of the mission.

The primary goals of thermal management are to:

- Keep spacecraft components like bus/payload/battery within temperature limits. This requires identification of factors, which affect the component temperatures and manage these factors.
- Support the structural integrity of the spacecraft considering all external conditions, that is, external panels facing the Sun, and facing deep space.
- Determine the effects of a short-term temperature excursion, which go beyond of design limits on component integrity and performance.
- Provide redundancy in thermal management.

The spacecraft industry has developed a systematic approach to the design of thermal subsystems for large spacecraft. Following best practices of the spacecraft industry, the thermal design process for CubeSats should consist of several steps (Fig. 2). If a CubeSat designer will follow the proposed approach, many mistakes in the design and potential mission failure could be prevented.

2.1 Requirements

A first step in good engineering design should be the establishment of requirements. Before starting a CubeSat design, the following information should be available upon which the design requirements will be based:

- Mission parameters: design life, orbit, vehicle operations, safe mode, mass limits, and deorbiting;
- Thermal parameters: amount of waste heat, temperature limits, and material criteria.

The worst-case scenario in waste heat generation and dissipation, environments (e.g., sun loading), operating modes and contamination/degradation of radiator surfaces, etc. from launch to deorbiting should be considered.

2.2 Conceptual design

The conceptual design process consists of two steps. First a designer should consider *conceptual design options* like aspects of the CubeSat form factor (e.g., 1U and 2U, etc.), solar array location (e.g., body mounted or deployable), radiator design (e.g., location of body mounted, material, and surface finish), multilayer insulation (MLI) (e.g., type and size), phase change material (PCM) (e.g., type and melting

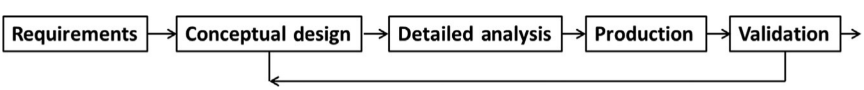

Fig. 2 Workflow of design of thermal system.

temperature), and louvers [3]. Secondly a *preliminary analysis* should be undertaken. It should include steady-state heat balance estimation at "a back-of-the-envelope" level. Hot and cold cases should be considered based on the mission requirements, orbit conditions (Section 2.1), and options selected from the aforementioned. The analysis conducted should answer a basic question: for given requirements and chosen options, will all thermal requirements met? If the answer is NO, the designer should reconsider options and find the ones (e.g., do we need heat pipes, PCM, and deployable radiator?) that meet all of the thermal requirements.

2.3 Detailed analysis

After completion of the conceptual design, a detailed analysis should be conducted to determine if the CubeSat design meets all requirements. It includes building a spacecraft thermal model that should take into account all important components, thermal links, orbits, etc. This thermal model predicts temperatures at the interface between components and the bus frame under worst-case hot and cold conditions. Sometimes a detailed thermal model at the component level is needed, in particular, for components with significant heat discharge.

2.4 Design validation

The thermal control system design is validated during thermal vacuum testing (TVAC testing). Test procedures for TVAC tests are prepared using the thermal model previously developed (Section 2.3). The TVAC test validates the thermal model and the thermal control system design, if successful. The model is also used to determine ideal positions of test thermocouples during TVAC testing. The accuracy of the thermal model is determined by comparison of the simulation results with the thermal vacuum test.

3 Thermal management challenges

The simplest way to remove waste heat from components is to mount heat dissipating components on a radiator inner surface. The waste heat will spread over the panel by conduction or with help of heat pipes and dissipate into space via radiation. However, CubeSat architecture often forces designers to mount boxes internally on a spacecraft frame, away from the radiators.

Previously, radiators have not been used widely on CubeSats. Waste heat has been traditionally rejected from components into the space from component surfaces. This strategy has been relatively effective because not much waste heat has been generated and the components were distributed sparsely so each component had a direct view to space. However, most component surfaces have not been treated to serve as a radiator so the efficiency of heat rejection is not high utilizing this strategy.

With the increasing complexity of CubeSat missions and corresponding CubeSat designs, the packing factor of components in the new generation of CubeSats is dense.

Components block each other so their view factor to space is minimal. Under such circumstances, radiators must be used for waste heat rejection into the space. This creates a need to transfer waste heat from the components to the radiators. Both mechanisms of heat transfer, conduction and radiation, transfer waste heat from components to radiators. The radiation portion can be enhanced by increasing emissivity of both, the internal and external surfaces and increasing radiator temperature. This can significantly contribute of total heat transfer, for example, up to 20%–30% [4]. For high fluxes of waste heat, heat pipes could be needed also to enhance waste heat transfer from components to the radiators and spreading heat over radiator surface.

Typical CubeSat systems that present thermal management challenges include the following:

- Batteries—usually operate in smaller temperature range than other electronics and lower operating temperatures. The best way to achieve effective thermal control of batteries is to have a separate thermal control system with dedicated radiator and heater.
- Optical elements—do not tolerate large temperature changes. Large temperature gradients across the optics should be also avoided to prevent thermal distortion.
- IR detectors—operate at very cold temperatures and typically require cryogenic systems. Complications might come from very close proximity of components operating at high temperatures (e.g., electrical propulsion) to IR detectors.
- Radiators—as the data in Table 1 indicates, rejection of waste heat from CubeSats require much higher density than from GEO communication satellites. Essentially, the required heat flux is above a typical range of heat rejection like 150–300 W/m^2. There are two possible solutions to this problem:
  - Increase radiator temperature. This can be done only if different radiators will be assigned to different heat generating components. This allows increasing radiator temperature for some components and reducing the radiator size. For example, some electronics can work at 70°C, while battery temperatures cannot exceed 40°C. If both, electronics and batteries, are connected to the same radiator, the radiator size is determined by sum of fluxes of waste heat from both sources and the maximum allowable temperature for the battery. However, if electronics and the batteries are connected to different radiators, a total radiator area would be less than in the first case.
  - Use a deployable radiator that increases radiator area while keeping the spacecraft size constant. However, use of the deployable radiator necessitates a solution of a very complicated technical problem: transferring waste heat from the spacecraft body into the deployable radiator across the hinge with a minimal temperature drop across the hinge. The loop heat pipe (LHP) represents one of the possible solutions. However, the LHP brings its own set of problems.

4 Heat balance estimation

At the conceptual design step (Section 2.2), it is paramount to conduct a preliminary analysis based on steady-state heat balance estimation for extreme thermal environments (hot and cold cases) that will drive the spacecraft design. The thermal environments depend on orbit and spacecraft parameters including spacecraft dimensions and power dissipation. The operating environments are described in the succeeding text:

I. Low Earth orbit (LEO) missions are between 400 and 800 km altitude. Albedo and Earth infrared loads significantly contribute into the thermal environment and must be taken into account. Typically a period of a circular orbit is around 90–95 min.

II. Geostationary (GEO) missions generally maintain a constant altitude. Albedo and Earth infrared contribution into thermal environment is small.

III. High Earth orbits can be highly eccentric with a very low perigee (several hundreds of kilometers) and with an apogee of many thousands of kilometers. The thermal environment changes from significant influence of Albedo and Earth IR at the perigee to only direct solar contribution at the apogee.

IV. Interplanetary missions can have a significant variation in solar flux depending on the distance to the Sun.

The major task of a preliminary analysis is to determine thermal balance of the radiators. This will affect the spacecraft design in a significant way. A rough estimation of required radiator size and optical treatments will be determined based on the thermal balance equation. Fig. 3 shows a hot case thermal environment for a CubeSat: the spacecraft is exposed to solar, Albedo and Earth IR fluxes. The absorbed heat is radiated to space through radiators. Under these assumptions, the energy balance for a radiator can be written as follows:

$$Q_{ext} + Q_{int} + Q_{backload} = Q_{rad} \tag{1}$$

where Q_{ext} is the external heat (e.g., from the Sun) absorbed by the radiator, Q_{int} is the internal heat generated by the satellite, $Q_{backload}$ is the external heat load on other surfaces, and Q_{rad} is the heat radiated by radiator.

Substituting Eq. (1) for

$$Q_{ext} = q_{ext}A \text{ and } Q_{rad} = \varepsilon\sigma AT^4$$

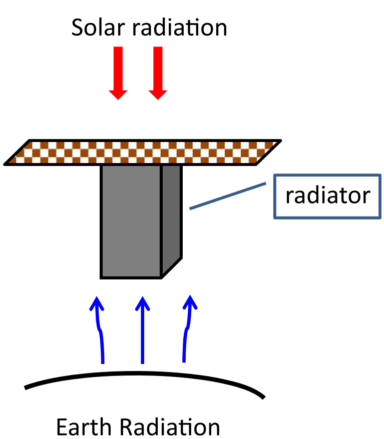

Fig. 3 Thermal environment.

The radiator heat balance becomes

$$q_{ext}A + Q_{int} + Q_{backload} = \varepsilon\sigma AT^4 \tag{2}$$

where q_{ext} is the external heat load on the radiator per unit area, A is the radiator area, ε is the radiator emissivity, σ is the Stefan-Boltzmann constant (5.67×10^{-8} W/m^2 K) and T is the radiator temperature.

The external heat load on the radiator can be split on four components:

$$q_{ext} = q_{solar} + q_{albedo} + q_{EarthIR} \tag{3}$$

where q_{solar} is the solar load per unit area, q_{albedo} is the albedo load per unit area, and $q_{EarthIR}$ is the Earth IR load per unit area. Typically, $q_{albedo} = 0.3 * q_{solar}$, and $q_{EarthIR}$ is calculated as the blackbody radiation at temperature of 250 K.

Heat balance Eqs. (2), (3) are used for consideration of two extreme, hot and cold, cases to design a spacecraft that meets the temperature requirements on both upper and lower limits. The power profile for the hot case corresponds to the case in which all components are at the highest level of heat dissipation, while the orbit is such that the spacecraft is exposed to maximum solar, albedo and Earth IR loads. All margins should be included into the input data to produce the maximum possible temperature. Similarly the input data for the cold case should be selected to result in the lowest possible temperature. However, the assumptions made for both cases should be realistic and correspond to the considered mission. For example, solar flux cannot be perpendicular to both, radiator and solar array, if they are perpendicular to each other.

A preliminary analysis always begins with consideration of the hot case. In the hot case a radiator area is estimated for a given external heat load, a given internal heat generation level, and a given radiator temperature. The estimated radiator area should be compared with an available external surface of the spacecraft for heat rejection into space. In the cold case, radiator temperature is determined for a given radiator area (determined earlier in the hot case study) and the lowest possible heat load including external and internal heat loads.

Both cases are very important for a preliminary analysis. The hot case determines the minimum required size of the radiator area. If the analysis shows that the required radiator area is larger than the available CubeSat external surface, something must be done to resolve the problem. Possible solutions include the following:

(a) Increase radiator temperature.
(b) Use a deployable radiator.
(c) Accumulate waste heat at the peak demand and reject heat later.
(d) Increase spacecraft size, that is, add section(s) to satisfy requirement for radiator area.

The solutions (a) and (b) have been discussed in Section 3. The solution (c) is possible when generation of waste heat varies with time. The heat is stored at the time of peak generation and rejected into space at the time of low demand. The heat accumulator could be a component with high heat capacity or phase change material (PCM). The later uses a latent heat of melting/solidifying to store and release heat at the constant temperature.

The cold case defines the coldest spacecraft temperature under given thermal conditions when a spacecraft radiator is defined per the hot case. If the lowest temperature is below allowable temperature, heaters must be used to raise component temperature above the lowest limit. If a PCM-based accumulator is used for temperature fluctuation moderation, latent heat of solidification must be included in the heat source contributions.

4.1 Example

Problem: Determine the required radiator area and heater size (if mandatory) for the 6U CubeSat depicted in Fig. 4. The component heat dissipation is 50 W. The spacecraft has deployable solar arrays. Assuming an isothermal radiator and energy balance governed by Eqs. (2), (3), the radiator area that will keep temperature of the radiator below or equal to T_{max}, is determined to be

$$A = \frac{Q_{int} + Q_{backload}}{\varepsilon\sigma T_{max}^4 - (q_{solar} + q_{albedo} + q_{EarthIR})} \tag{4}$$

Assume heat generation: $Q_{int} = 50$ W (hot case); $T_{\min} = 273$ K; $T_{\max} = 313$ K; $q_{solar} = 1420\,\text{W/m}^2$; $q_{albedo} = 0.3^* q_{solar} = 426\,\text{W/m}^2$; and $q_{EarthIR} = 240\,\text{W/m}^2$. Optical properties of external surfaces are as follows: emissivity (ε) -0.8, solar absorptivity (α) -0.2; orbit parameters, inclination 0 degrees, altitude—500 km; view factors for ±Y and ±X panels to Earth is 0; and view factors ±Y and ±X panes to Space is 1.0.

4.1.1 Hot case

We will explore which of the two positions, (a) and (b), depicted in Fig. 4 constitutes the worst case.

As one can see from Table 2, position (b) is the most challenging, because the required radiator area is large than the available external surfaces available for radiation, that is, there is not enough surface to keep the radiator temperature below 313 K. Calculations show that an available radiator area of 0.14 m^2 can keep the radiator

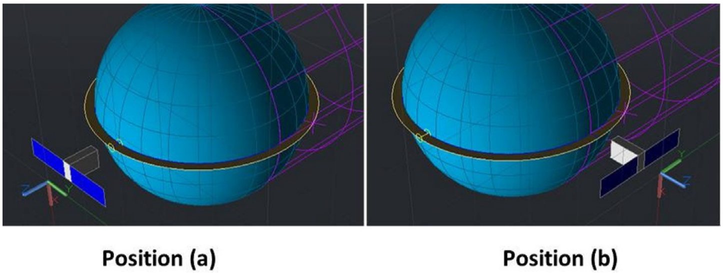

Fig. 4 Example.

Table 2 Hot case study.

	Position (a)		Position (b)	
$+Z$ Panel 0.02 m^2	Receives solar heat load Q_{+Z}	$Q_{+Z} = 0.02\ {}^*\ 0.2\ {}^*\ 1420 = 5.68$ W	Radiates heat into space	
$-Z$ Panel 0.02 m^2	Receives albedo and Earth IR heat loads	$Q_{albedo} = 0.3\ {}^*\ 5.68 = 1.74$ W $Q_{Earth\ IR} = 0.02\ {}^*\ 240 = 4.8$ W	Receives albedo and Earth IR heat loads	$Q_{albedo} = 0.3\ {}^*5.68 = 1.74$ W $Q_{Earth\ IR} = 0.02\ {}^*\ 240 = 4.8$ W
$+Y$ Panels 0.06 m^2	Radiates heat into space		Radiates heat into space	
$-Y$ Panels 0.06 m^2	Radiates heat into space		Receives solar heat load Q_{-Y}	$Q_{-Y} = 0.06\ {}^*\ 0.2\ {}^*\ 1420 = 17.04$ W
$\pm X$ Panels 0.03 m^2	Radiates heat into space		Radiates heat into space	
Backload	$Q_{backload} = 5.68 + 1.74 + 4.8 = 12.22$ W		$Q_{backload} = 17.04 + 1.74 + 4.8 = 23.58$ W	
Available radiator [m^2]	$2\ {}^*\ 0.06 + 2\ {}^*\ 0.03 = 0.18$		$0.02 + 0.06 + 2\ {}^*\ 0.03 = 0.14$	
Required radiator [m^2]	$A = \frac{55 + 12.22}{0.8 * 5.68 \times 10^{-8} (313)^4} = 0.14$		$A = \frac{55 + 23.58}{0.8 * 5.68 \times 10^{-8} (313)^4} = 0.17$	

temperature equal or below 328 K, that is, 15 K higher than T_{max}. Another option would be to use 12U satellite to increase area of ±X panels.

4.1.2 Cold case

The spacecraft is in an eclipse; maximum internal heat generation is assumed to be 10 W; the minimum radiator temperature T_{min} should be equal to or higher than 273 K (0°C); all panels with exception of the $-Z$ panel radiate heat into space. The spacecraft temperature is determined from heat balance Eqs. (2), (3):

$$T = \sqrt[4]{\frac{Q_{int} + Q_{backload}}{\varepsilon \sigma A}}$$

Table 3 shows that the radiator temperature would be around 200 K (−73°C) in the cold case, much lower than T_{min}. Additional heater(s) are required to bring the spacecraft radiator temperature to a reasonable level. For example, an additional 35.5 W heater is required to bring the radiator temperature to 0°C. To meet such a requirement

Table 3 Cold case.

	Eclipse	
$+Z$ Panel 0.02 m^2	Radiates heat into space	$Q_{Earth\ IR} = 0.02\ ^{*}\ 240 = 4.8$ W
$-Z$ Panel 0.02 m^2	Receives Earth IR heat loads	
$\pm Y$ Panels 0.06 m^2	Radiates heat into space	
$\pm X$ Panels 0.03 m^2	Radiates heat into space	
Available radiator [m^2]	$2\ ^{*}\ 0.06 + 2\ ^{*}\ 0.03 + 0.02 = 0.2$	
Radiator temperature [K]	$T = \sqrt[4]{\frac{10+4.8}{0.8{*}0.2{*}5.68x10^{-8}}} = 200$	

could be a challenge because it will require the use of a battery during eclipse when battery charging is not possible. This means that the results of cold case analysis affect battery sizing.

5 Power

A CubeSat designer should be aware that the capacity of waste heat rejection for a given radiator can limit energy generation by the spacecraft energy sources (solar arrays), specifically the ability for the spacecraft thermal control system to dissipate the residual heat associated with the power source. As an illustration, consider a problem from Section 4.1 assuming 60% efficiency of power utilization. This means that the total power generated by the solar arrays is 125 W. If the power level generation surpasses 125 W, the waste heat generation will exceed 50 W, which will lead to an increase in the radiator temperature. Such an increase will affect component temperatures.

Maximum energy production for different CubeSats was investigated in previous studies by Yender [1] (see Fig. 5). The study assumes that spacecraft has solar deployable arrays and energy efficiency of 60%. The study has shown also a strong effect of optical properties of external surfaces on upper limit of energy production. Higher solar absorptivity leads to low energy production limit. On the other hand, higher emissivity facilitates higher energy generation limits.

6 Hardware for satellite temperature control (STC)

The goal of the STC system is to keep the temperature of all satellite components within limits. The STC employs many different types of hardware to achieve this goal. Heaters, multilayer insulation (MLI), heat pipes, finishes of surfaces, phase change materials, etc. are employed by the STC. A detailed description of the hardware that is used in large satellites can be found elsewhere [5]. We will concentrate on use of the hardware in CubeSats.

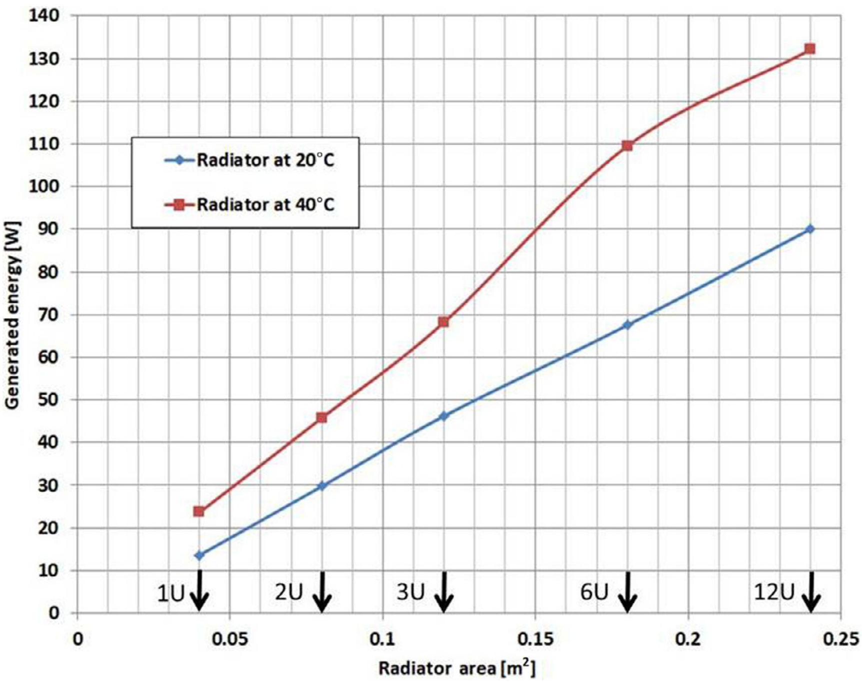

Fig. 5 Useful energy generation for CubeSats [1].
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6.1 Multilayer insulation (MLI)

MLI is typically made of multiple layers of metallized plastic film conductively isolated from each other by spacers and assembled as blankets. Externally, MLI reduces thermal radiation to space and protects against excessive heat fluxes such as Sun and thruster plume heating. Internally, MLI blankets are used to create areas of separate thermal control, to reduce temperature gradients across the spacecraft, and to reduce heater power consumption. Blankets have to be grounded to the spacecraft structure to prevent electrostatic charge buildup and arching. MLI blankets have to have vents to reduce outgas pressure to minimize gaseous conduction effects. Due to fact that both radiative and conductive heat transfer occur simultaneously, the best way to describe heat transfer through MLI is to use coefficient of effective emissivity (e* or ESTAR) which is defined as

$$Q_{net} = A_{MLI} \varepsilon^* \sigma * \left(T_{in}^4 - T_{ex}^4\right)$$

where A_{MLI} is MLI area, ε^* is effective emissivity (Estar), and T_{in} and T_{ex} are temperatures of the inner and out layers.

CubeSat designers should be aware that small patches of MLI show a significant variation of properties across their surface. For example, Estar for 20 layers MLI blanket (12 in × 12 in) changes from 0.006 in the middle to 0.15 close to seam, which indicates a change of heat loss from 3 w/m^2 to 30 w/m^2 [5]. The situation will be even worse for smaller MLI blankets.

6.2 Highly conductive strips

Typically, components inside a CubeSat are mounted on shelves that are attached to the space frame. Heat that is generated by a component has to flow from the component through fastener to the shelf, along the shelf, through fastener of the shelf to the CubeSat frame and from the frame to the radiator. Rough estimations indicate that the thermal conductivity of such a pass is about 0.5 W/K. If the amount of the transferred heat is a couple of W, a temperature drop across the pass is not significant. However, transfer of 20 W would require a temperature difference of 40°C. Assuming the maximum component temperature of 60°C, the radiator temperature has to be at 20°C. This requirement will lead to a significant increase in the radiator size. Creation of a thermal path with minimum resistance is very important for CubeSats with power above 20–30 W. Thermal resistance of fasteners like screws and bolts can be very significant [5] and must be taken into account during design of the thermal control system.

Using a copper strip could help but only for a small heat flux. For example, transferring 10 W from the middle of the CubeSat to the surface at a 3°C temperature difference would require the cross section of the strip be about 4 cm^2, which is quite significant. A heat pipe is the best solution for transferring more than 10 W of heat across these types of junctures.

6.3 Heat pipes

Heat pipes could be useful for several applications. Among them are to:

- Conduct heat from a component mounted on a shelf to a CubeSat external surface.
- Transfer solar load from a sunny side to a shaded side. For example transfer heat from −Y panel to +Y panel for spacecraft in position (b) in Fig. 4. The heat pipes are located on the external surfaces and connected to each other.
- Connect an external panel with a deployable radiator.
- Spread heat over radiator to reduce temperature gradients along radiator surface.

There are several requirements associated with the use of a heat pipe including the following:

(a) Heat transfer liquid and material of the heat pipe should be compatible with each other. Previous work extensively discusses this subject [5].
(b) Special attention should be paid to the position of the heat pipes relatively to the Earth during TVAC test, namely, heat pipes should be horizontal to the surface of the Earth.

6.4 Thermal surface finishes

An increase of CubeSat power intensifies the radiation heat exchanger inside the spacecraft and heat rejection into space. For external surfaces, optical solar reflector

(OSR), white paints and other treatments are used to minimize absorbed solar energy while maximizing heat rejection. If solar arrays are mounted on the spacecraft, the area of the arrays should be treated as having high solar absorptivity ($\alpha \approx 0.75$–0.8) and high IR emissivity ($\varepsilon \approx 0.85$–0.9). Black paint is commonly used on internal surfaces of the spacecraft to enhance radiative heat transfer among internal components and radiative exchange with the internal surfaces of radiators.

6.5 Phase change material (PCM)

PCM absorbs energy during melting and releases the energy during solidification. Both, melting and solidification, occur at the same temperature. The use of PCM-based systems for thermal control is not new [5]. A general application of PCM thermal control is for cyclically operating components that are operating on an ON-OFF cycle. The same is true for environment heating of CubeSats. For example, panels of the spacecraft shown in Fig. 4 are periodically exposed either to a Sun load or are facing space. A PCM accumulator can store solar energy when the radiator is facing the Sun and release solar energy during shadow. Storing energy at the peak load and releasing it later significantly reduces required radiator size.

PCMs are differentiated by melting point and latent heat of fusion. Extensive description of PCMs can be found in Refs. [4,5]. Some PCMs like Octadecane or Eicosane have melting point temperatures around 40–60°C, which makes them attractive for thermal control in CubeSats. Paraffin Waxes could be used for CubeSat thermal control systems [6,7]. The toxicity of PCMs should be taken into account, in particular, for CubeSats deployed from the International Space Station due to NASA safety concerns.

Designers of PCM accumulators face two major problems, namely, low heat conductivity of a PCM and changes in PCM volume during the phase change. A solution to the first problem could be the use of fins/fillers to provide low thermal resistance paths through the PCM [5,7]. The container for PCM should be leak-proof and flexible to accommodate the volume change during melt or freezing PCM. It is reported [8] that carbon velvet thermal interfaces are able to accommodate PCM volume change.

6.6 Telemetry and commands

A CubeSat should have a sufficient number of temperature sensors and channels to determine the status of critical components and efficiently control heaters from ground. Using an on-board computer for the thermal control system reduces number of telemetry channels that are dedicated to temperature sensors. Typically, temperature sensors are installed on components that

- generate large amounts of heat,
- operate at cryo-level temperatures,
- might have boiling or freezing problems,
- must operate in a narrow temperature range,
- should have a specific temperature difference with other component,
- are sensitive to temperature, like battery.

Flight temperature sensors should be calibrated (typically, during thermal-vacuum testing) over the temperature range that exceeds the expected range of temperature change of a component. It should be noted also that analog-to-digital conversion of temperature reading on the satellite and reverse digital-to-analog conversion on a ground station can lead to a significant loss of accuracy of temperature reading.

A CubeSat can have thermal controls divided into several zones. For example, payload and bus can have separate temperature control zones. Another option would be to group components with similar temperature requirements as one thermal control zone. Components from the same group can be connected to one radiator, which could have the highest temperature and consequently the smallest size. It will reduce complexity of the design and improve thermal control efficiency.

Temperature sensors that are placed inside a component provide information on performance and health of the component. However, additional temperature sensors should be installed to monitor performance and health of the CubeSat. Location and parameters of such temperature sensors should be determined by simulations and verified during a thermal vacuum (TVAC) test.

Temperatures during TVAC tests are monitored by thermocouples placed in areas where the temperatures are indicative to the thermal performance and may be used to make comparisons with simulation results. The thermocouples are usually mounted using adhesive tape to avoid damaging the thermal finish. Positions of test thermocouples are shown in Table 4.

A more comprehensive description of the thermocouple locations at the TVAC test can be found in Ref. [9].

If a CubeSat is designed for LEO and CubeSat crosses shadow/light separation line twice during each 90 min, it might put an additional thermal stress on the CubeSat structure. Existing TVAC chambers cannot simulate such rapid switch from hot to

Table 4 Thermocouples locations at the TVAC test.

Item	Thermocouple location	Goal
Important electronics component (like battery)	Component baseplate	Verify calibration of flight thermistors
Frame	Near flight thermistors	Verify calibration of flight thermistors
Heaters	Near heater	Track heater activity and verify performance
Thermostats (if applicable)	Near thermostats	Verify thermostats set points
Mounting platform and radiators	Corresponding to nodalization of thermal model	Comparison of thermal model and TVAC test results

cold. Specially designed thermal shock TVAC chambers should be used for such a test. The importance of TVAC testing to validate the thermal models developed for a mission cannot be overstated.

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